Crosslingual learning in multilingual text detoxification

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Abstract:
Online toxicity presents increasing challenges for content moderation, especially in multilingual settings where cultural nuance and grammatical variation complicate detection. This study explores crosslingual fine-tuning and syntactic awareness as strategies for improving multilingual text detoxification—the task of rewriting toxic utterances into fluent, non-toxic equivalents while preserving meaning. We fine-tune the Qwen3-14B language model using LoRA adapters on parallel data from the PAN 2025 TextDetox challenge, combining English, German, and Russian. We evaluate three setups – English-only, token-supervised, and multilingual – across six target languages using official PAN metrics: Style Transfer Accuracy, Semantic Similarity, and Fluency, merged into a unified J-score. Results show that multilingual fine-tuning improves generalisation, outperforming baselines in most languages and unexpectedly achieving the highest detoxification quality in Ukrainian, despite no direct exposure during training. In contrast, token-level supervision offers limited gains. Further analysis reveals model difficulties with complex morphology, indirect forms of toxicity, and cultural references. We conclude that crosslingual transfer is a promising avenue for improving detoxification in low-resource languages, though future systems must better integrate syntactic cues and contextual understanding to address subtler forms of toxicity.

1 Introduction

Abusive and toxic language in online public spaces remains a significant issue despite regulations and efforts from governments and social media platforms. Anti-social behaviour has only increased with internet penetration [1]: online sexism is on the rise [2]; in 2021, more than half of Swedes between 16-45 reported having been targets of online hate speech and threats [3]. Socially vulnerable groups are especially at risk, with minors particularly likely to lack the resilience to withstand such attacks.

Toxic communication has serious consequences at both societal and individual levels, threatening the mental wellbeing of its targets, with potentially serious or life-threatening psychological consequences [3, 4]. It can also lead to the disengagement of targeted groups from public discourse, thereby reducing the diversity and quality of online conversations [3].

Platforms continue to struggle to balance content moderation and freedom of expression [2]. Most have adopted a reactive deletion strategy rather than a proactive detection and mitigation strategy, allowing hate, misinformation and negative stereotypes to propagate unchecked. Manual intervention is simply not feasible or effective on the massive scale required to police online public spaces in real time. To address multilingual moderation, platforms currently rely on machine translation (MT) to translate content on their sites [5]. However, MT often fails to render low-resource languages and cultural nuance accurately [6], raising questions as to the practical effectiveness of this strategy – and further underscoring the importance of multilingual detoxification strategies.

However, multilingual text detoxification remains a significant challenge [7]. Most datasets and efforts to address this issue have centred on English [8, 9], despite only a quarter of the world's internet users being native English speakers and speakers of the most-used languages on the internet together only accounting for three-quarters of users [10]. Small datasets can lead to overfitting issues due to too little linguistic variation [11]. While efforts have been made to alleviate data sparseness by creating synthetic data through backtranslation [3], results have been inconsistent across languages [8].

To reduce digital violence and create more civil, productive online spaces in the future – in both human and Large Language Model (LLM) communications – it is vital to work towards effective automatic and multilingual text detoxification strategies. These have the potential to mitigate the harms of cyberbullying, hate speech and societal polarisation – while also reducing the burden on human moderators [11]. However, developing effective text detoxification systems presents several challenges: toxicity is subjective, culturally variable and context-sensitive. Systems must strike a balance between editing harmful content and preserving meaning, fluency and free expression.

One initiative driving progress in this area is the Conference and Labs of the Evaluation Forum (CLEF), an organisation promoting research and development in multilingual and multimodal information systems. Their PAN 2025 multilingual text detoxification challenge [12] provides a parallel dataset across 15 languages to support the development of strategies in this field.

1.1 Research questions

Two questions remain underexplored:

- Can crosslingual fine-tuning improve detoxification in unseen or low-resource languages?
- Can token-level supervision make detoxification more accurate?

In this work, we utilise the dataset from the PAN 2025 challenge to fine-tune *Qwen3-14B*, a state-of-the-art multilingual decoder-only model, chosen for its strong zero-shot crosslingual abilities, and apply parameter-efficient fine-tuning with LoRA to adapt it to the detoxification task. The task is to identify and edit instances of toxic content so that they are no longer toxic, while saving as much of the semantic content of the source text as possible. Outputs are evaluated via the metrics associated with the PAN 2025 shared task: Style Transfer Accuracy (removal of toxic content), Semantic Similarity (semantic content preservation) and Fluency. These metrics are crucial because detoxified output must be acceptable to human users, avoiding semantic drift, hallucinations or grammatical mistakes, otherwise risking user trust and acceptance. We present both quantitative results and qualitative analyses across English, German, and Russian, including on language-generalisation to unseen languages. Rather than optimising for

leaderboard performance, our goal is to explore the interpretability and crosslingual generalisability of detoxification systems.

1.2 Hypotheses

1. Multilingual fine-tuning will improve detoxification quality across languages.
2. Token-level toxicity supervision will improve detoxification.

We hypothesise that the model's pre-trained crosslingual representations will support generalisation to other languages. We also explore whether conditioning on token-level toxicity annotations can help the model focus more effectively on syntactic and semantic cues — enabling more accurate, controlled edits of toxic language. This offers new insights into crosslingual generalisation and conditioning for LLM-based moderation tools.

1.3 Paper organisation

The remainder of this paper is organised as follows: Section 2 reviews the state of the art in multilingual text detoxification. Section 3 describes our methodology in detail. Section 4 presents our experimental results and discusses the findings. Section 5 reviews the conclusions and offers directions for future research. Finally, Section 6 answers the questions set in the exam paper.

2 Background

2.1 Detecting toxicity

A clear and agreed-upon understanding of toxicity is vital for the development of fair and interpretable detoxification models and systems. While there is no universally agreed-upon definition of toxic communication in the NLP community, it is typically characterised as communication that:

- is rude, disrespectful, pejorative, unreasonable, abusive, offensive or demeaning;
- uses vulgar or profane expressions;
- directs hate or insults towards individuals or groups, often based on protected characteristics such as race or gender [4, 8, 3, 2, 13].

Importantly, toxicity is not equivalent to negative sentiment [14]. A key subcategory of toxic communication is "hate speech", which incites violence or hatred specifically towards a protected group and is thus explicitly illegal in some jurisdictions,[1] whereas abusive language in general is not.

Toxicity can be explicit – involving direct or overt attacks – or implicit – involving indirect or coded discrimination, stereotyping, prejudice or bias. The latter is much more challenging for models to detect due to more frequent complex, indirect and figurative language, such as sarcasm or humour [9]. A range of categorical distinctions for toxicity have been proposed based on factors such as offensiveness intensity, topic and targeted entity [11, 9]. Definitions also vary across datasets, with some using binary labels (i.e., (non-)toxic), others multiclass labels (e.g., hate, insult, threat), and some focusing only on explicit vulgarity. The PAN detoxification challenge limits its scope to explicit toxicity – i.e. vulgar or profane language targeted at individuals or groups – due to the challenges posed by implicit toxicity [15]. We therefore adopt the same scope in this study, although we recognise that this simplification does not adequately capture the true spectrum of toxic communication.

Annotations of toxicity are highly subjective and inter-annotator agreement is often low, even when training and guidance is provided [11], presenting challenges for both training and evaluation. This issue is amplified in multilingual contexts, where cultural understandings of offence may diverge – simultaneously highlighting the constructed nature of the concept of toxicity as a concept.

Our collective understanding of toxicity is shaped by social media platforms, whose practices are mediated by their terms of use and potential commercial interests [11]. For instance, Meta defines "hateful conduct" as attacks based on protected characteristics or membership to a vulnerable group. It removes content including dehumanising speech, harmful stereotypes, criminal allegations, swearing and slurs, serious insults, expressions of contempt or disgust and calls for segregation or harm.[2] Meta further notes that it makes considerations "based on local nuance" – highlighting a key challenge in multilingual text detoxification: navigating cultural and linguistic variability in what is considered toxic (see Section 6.2).

Because platforms often define toxicity in alignment with dominant cultural norms, relying exclusively on such definitions risks reinforcing a narrow or culturally biased conception of toxic language. Meta and X may perpetuate an anglophone-centred conception. Meanwhile, in 2024, Weibo started removing posts "showing off wealth and worshipping money" for espousing values contrary to state ideology; a state-defined idea of toxicity [16]. This is highly limiting for the generalisability of monolingual models (see Section 6.2).

2.2 Approaches to detoxification

A number of approaches have been proposed to tackle online toxicity. Some methods avoid altering the text at all, with counter-speech or counternarratives instead generating a non-aggressive response to the toxic content intended to counteract its impact [17, 18, 9]. In contrast, text detoxification aims to mitigate the harmful effects of toxic language by rephrasing it while retaining as much of the semantic content as possible [9, 15]. This process mainly involves localised edits by either deleting or rephrasing toxic elements; exact strategies depend on the source language, with Russian, for instance, relying more on the rephrasing strategy, while German employs both equally [15]. Detoxification has been framed through a range of different subtasks:

1. *Style transfer*: changes the style from rude/toxic to neutral/non-toxic while preserving the core meaning [7, 19];
2. *Paraphrasing*: rewrites the sentence while preserving semantic equivalence with either rule-based substitutions or seq2seq neural approaches [20, 21];
3. *Normalisation*: reduces the level of hate to make the content less likely to garner engagement [1];
4. *Translation*: Frames detoxification as a form of monolingual translation from toxic to non-toxic language or sentiment, with models learning to suppress harmful meanings while preserving informative content [22, 20].

[1] e.g., Denmark, *Straffeloven, §266B*; the UK, *Public Order Act 1986*.

[2] https://transparency.meta.com/en-gb/policies/community-standards/hateful-conduct/

5. Fine-tuning: Adapts a large pre-trained model, such as Qwen3, to directly generate detoxified text using a corpus of aligned toxic/non-toxic sentence pairs [21].

Different approaches are framed under various combinations of these subtasks, for example [23] conceptualise detoxification as a combination of paraphrasing and formality-oriented style transfer, while fine-tuned models learn such transformations implicitly. In practice, these approaches represent different perspectives on transforming a sentence – whether framing detoxification as a shift in style, language (translation) or meaning (paraphrasing) – and are typically implemented using either encoder–decoder architectures or decoder-only generative models.

2.3 Multilingual detoxification

Most detoxification research has focused on English, with far fewer resources available for other languages. In 2020, [11] reported that English-language datasets outnumbered those in other languages by a factor of five. This imbalance has persisted, even as interest in multilingual detoxification has grown.

Some have endeavoured to address the data sparseness problem through a range of methods, from synthetic data augmentation to back- or pivot-translation, which are prone to error propagation; others have explored sharing representations within the encoder or decoder [24].

The 2022 Russian-specific detoxification challenge demonstrated that different approaches excel in different subcomponents of the task [14]. For instance, combining a toxicity classifier with BERT-based token substitution preserved semantic content most effectively, while another system trained to conduct Levenshtein-based token replacement performed best in terms of style transfer. This highlights the importance of tailoring detoxification approaches to evaluation priorities.

Unsupervised approaches do not require parallel data, but often produce lower-quality outputs [14]. To date, the best results – including in the inaugural PAN detoxification challenge – have been achieved through the language-specific fine-tuning of large pre-trained models on parallel corpora of toxic–neutral sentence pairs [14, 15]. However, high-quality parallel datasets remain scarce [7], limiting the scalability of these methods across low-resource languages. This has prompted a growing interest in the potential of crosslingual knowledge transfer, a well-established technique in machine translation [25]. While it seems that language-specific fine-tuning outperforms zero-shot or transfer methods, the latter may be essential for enabling detoxification in truly low-resource languages.

Furthermore, a number of promising efforts have been made to equip models with additional sociolinguistic and morphosyntactic information in order to improve performance, notably in the field of machine translation [26, 27], including specifically in LLM prompts [28] and morphosyntax for synonym detection in Arabic [29]. Since detoxification can be framed as a monolingual translation task, enriching the input with linguistic or sociocultural cues may further improve model effectiveness.

2.3.1 Crosslingual learning While crosslingual learning has shown promise in a number of other NLP domains to improve performance in low-resource languages, it remains relatively underexplored in the context of detoxification[30]. Recent work has applied it to toxic language detection, notably finding that culture-specific conceptions of toxicity hinder transferability [30, 31, 32]. Two factors identified as beneficial for effective crosslingual knowledge transfer are pre-training on a large multilingual corpus and avoiding the next sentence prediction learning objective during pre-training [33]. The *Qwen3-14B* model fits both of these criteria, having been pre-trained on 36T tokens using a causal language modelling objective. In contrast, models like LLaMA 3 were pre-trained on smaller datasets (15T tokens). Qwen therefore presents stronger potential for multilingual detoxification tasks.

3 Methods

There are many ways to define and approach the detoxification task (see Section 2.2). We initially defined the text detoxification task as a multilingual paraphrasing problem. In order to achieve the best possible paraphrase, we followed diverse approaches in the beginning before we were able to condense it to the most promising methods.

Our initial idea was to build an ensemble architecture that would aim for minimal computational efforts: first, we would apply simple methods before consecutively employing more advanced and computationally expensive techniques, should the simpler methods not yield viable results. Our initial processing pipeline consisted of the following steps:

1. *Identify toxic word or span* based on phrase/dependency structure, since deleting a single word often leads to influent results.
2. *Delete toxic content* if it was not vital to meaning (such as a head noun). Vocative noun phrases, for example, were a good target (see Section 6.1).
3. *Find synonym* for toxic word/phrase via lexical substitution methods if it is key to meaning.
4. *Conduct a grammar check* using an LLM, as these succeed at mimicking natural human language.

Even though the idea behind it was well-intended, it did not translate well into practice. This was mostly because almost any step in the pipeline demanded a final grammar check and rephrasing, which meant employing a fine-tuned LLM to achieve the best performance. Thus, we came to the conclusion that we should condense our approach to a style transfer fine-tuning problem.

Fine-tuning a large language model requires three key components, namely: a pre-trained model to fine-tune, a fine-tuning method, as well as training data. In this section, we will give a detailed overview of these three components and how we constructed a fine-tuning pipeline before discussing the evaluation methods for the fine-tuned model.

3.1 Model

We select the *Qwen3-14B* model (henceforth "Qwen3") as the foundation architecture of our fine-tuning using the *UnSloth* framework. Qwen3 is a transformer-based, decoder-only LLM developed by Alibaba Cloud, featuring 14 billion parameters [34]. This architecture enables strong generalisation and reasoning abilities, making it a suitable fit for complex natural language tasks, such as text detoxification.

Another deciding factor for this model was its multilingual pre-training on 36 trillion tokens across 119 languages and dialects (in-

cluding multilingual and domain-specific data such as code, maths, and instruction datasets) using a causal language modelling objective. Qwen3, therefore, presents stronger potential for the task at hand in comparison to models like LLaMA3 (trained on 15T tokens with a next-token prediction objective) (see Section 2.3.1). These multilingual capabilities enable crosslingual generalisation across diverse languages, meaning that fine-tuning the model on only a few languages could improve the overall detoxification performance in other languages as well, with promising implications for low-resource languages in real-world deployment.

3.1.1 LoRA Fine-tuning all 14 billion parameters of Qwen3, is not feasible on the computer hardware available to us. To reduce the computational cost of fine-tuning and avoid full weight updates, we employ the Low-Rank Adaptation (LoRA) method, introduced by [33], which is a more efficient implementation of adapter layers.

LoRA freezes the original model weights and injects trainable low-rank matrices into selected layers of the transformer [33]. It is commonly applied to the query, key, value, and projection layers of the attention mechanism [33], which are crucial for understanding context and relevance. However, recent work suggests that also adapting the feed-forward layers can improve performance [35], as these are responsible for deeper semantic abstraction. We decided to apply LoRA to both the attention mechanism and the feed-forward layers.

Our initial model had 14 billion parameters, which LoRA reduced to $\approx$ 128 million trainable parameters[3] (for more details on training parameters see Section 3.3).

Another advantage of not updating the initial model weights is that they can be quantized from high-precision 32-bit float values to lower-resolution 4-bit values, since high precision is only important when training the full weights. This reduces the model's size and allows for greater memory efficiency [36]. Previous research has shown that LoRA succeeds in fine-tuning quantized language models [37], making it a viable approach.

3.2 Data

3.2.1 Multilingual Paradetox In order to fine-tune our model to the multilingual detoxification task, we employed the Multilingual Paradetox dataset, provided and introduced by Dementieva et al. [38].[4] The dataset contains, as its name already suggests, multilingual data in nine different languages across different language families.[5] It is a parallel dataset consisting of a total of 3600 toxic–neutral sentence pairs (400 in each language) with an additional language tag.

The toxic sentences were drawn from different sources, depending on the language. The majority of samples are social media posts [7], with the German data, for example, based on three different social media comment datasets primarily about political events in Germany or the US [15]. The neutral sentences were created during the annotation process, with native speakers asked to formulate the neutral counterpart to each toxic sentence while following strict guidelines that aimed to ensure maximal content and fluency preservation, as well as toxicity neutralisation [38].[6]

As part of the PAN 2025 challenge, the same authors also provide test data for evaluation. This dataset contains 600 toxic sentences in each of the nine training languages, as well as six additional test languages to explore crosslingual knowledge transfer.[7] However, since the neutral sentences were not made public, we could not use this dataset for the evaluation of our models.

3.2.2 Data split Our analysis focuses on a range of typologically and culturally diverse languages. For the fine-tuning process, we extracted the English, Russian and German subsets of the Multilingual Paradetox dataset as these are our first or second languages and we can therefore perform fine-grained qualitative analysis on model outputs. We then deployed a 75/25 train-test split to draw the training and testing data from the same distribution.

To perform evaluation on the final model, we also select 25% of the Ukrainian, Spanish and Chinese subsets, next to English, German, and Russian. This selection allows us to test the crosslingual capacity of the model across languages that differ in morphological complexity, syntactic structure and cultural conceptions of toxicity.

3.3 Model training

To adapt the Qwen3 model to the task of multilingual text detoxification, we employ three supervised fine-tuning approaches using Google Colab with a Tesla T4 GPU. Each approach varies in the number of languages or the input format. To strike a balance between computational efficiency, stability, and performance, all models are fine-tuned for three epochs with a batch size of six, five warmup steps, a learning rate of $2e^{-4}$, and a 75/25 train-test split on each used dataset.

3.3.1 Multilingual Fine-Tuning (English, German, Russian) In our first setup, we fine-tune Qwen3 on a combined dataset containing English, German, and Russian data. The input format follows Qwen3's standard conversational structure [34], consisting of a system prompt paired with user input and assistant output. The system prompt includes a concise definition of toxicity, a step-by-step strategy to detoxify a given text, as well as several pairs of toxic and neutral sentences. The full system prompt can be found in the Appendix (A). The goal of this setup is to encourage crosslingual generalisation and robustness.

3.3.2 Monolingual Fine-Tuning (English) To assess the impact of multilingualism in the training data, we repeat the same procedure using only English examples. The format and structure of the data remain consistent with the multilingual setup: each instance includes a system prompt describing the task and a conversational output pair formatted according to Qwen3's chat interface. We can use this setup to explore whether crosslingual overlaps or token-level mismatches across languages have potential inference effects.

[3] In total, 128,450,560 trainable parameters.

[4] Available at: `https://huggingface.co/datasets/textdetox/multilingual_paradetox`.

[5] Training languages: English, Russian, Ukrainian, Spanish, German, Amharic, Arabic, Hindi, and Chinese

[6] `https://github.com/textdetox/multilingual_explainable_paradetox/tree/main/paradetox_collection`

[7] Additional test languages: Italian, Hinglish, French, Hebrew, Japanese, and Tatar

3.3.3 Monolingual Fine-Tuning with Toxic Token Annotations (English + Toxic Tokens) In a third variant, we expand the English-only training data by explicitly highlighting toxic tokens. This approach aims to support the model in identifying and targeting specific toxic words more effectively during the detoxification process. For each toxic input, we extract a list of toxic words using a pre-trained toxic-BERT classifier [39] that performed well on English data. The list of toxic tokens is included in the prompt to give the model explicit cues about potentially toxic content. As a result, the system prompt is slightly adjusted to include the token list for the input text and the example sentences. The full and adjusted system prompt can be found in the Appendix B.

3.4 Evaluation

To evaluate our models, we deploy the automatic evaluation pipeline provided by the CLEF 2025 TextDetox research challenge [12]. The evaluation setup is based on three features that reflect the core objective of text detoxification, namely reducing toxicity while keeping the meaning and linguistic quality:

Style Transfer Accuracy (STA) assesses whether the generated outputs are successfully detoxified. To enable robust evaluation across multiple languages, a separate fine-tuned *xlm-roberta-large* binary toxicity classifier is used.[8]

Content Similarity (SIM) quantifies how well the meaning of the original input is preserved in the detoxified output. For this, Language-agnostic BERT Sentence Encoder (LaBSE) embeddings are computed for the source sentence (in), generated target sentence (out_{gen}) and gold-standard human reference detoxification (out_{gold}) [40]. Next, the weighted sum of the output's cosine similarity between them is used as a measurement for semantic overlap. This enables the metric to account for paraphrasing and language variations while remaining sensitive to shifts in meaning.

Fluency (XCOMET) evaluates the linguistic quality of the rewritten text. The evaluation pipeline deploys $XCOMET - lite$, a state-of-the-art machine translation evaluation model, that reflects how likely the output is to be a fluent utterance in the target language [41].

To assess overall model performance, the three metrics are combined into a joint score (J) per sample as:

$$J = XCOMET \times (0.4 \times SIM_{in,out_{gen}} + 0.6 \times SIM_{out_{gold},out_{gen}}) \times STA,$$

with $XCOMET$, $SIM_{in,out_{gen}}$, $SIM_{out_{gold},out_{gen}}$, $STA \in [0, 1]$, (adopted from [15]).

4 Results & Discussion

In this section, we analyse and discuss the evaluation results of our three fine-tuned models and compare them to two different baselines. As baselines, we chose deletion of toxic keywords, as well as the backtranslation baseline, in which the toxic sentence gets translated into English before being detoxified with the *bart-base-detox* model,[9] and then translated back to the initial language. Both baselines were provided by the PAN challenge [15].

We tested the models not only on the languages they were trained on (English, German, and Russian), but also with three additional languages: Chinese, Spanish, and Ukrainian in order to explore possible signs of crosslingual transfer learning. We employ primarily the J-score (as presented in Section 3.4) in every language as a comparison metric. All our models were evaluated on the same test data. However, due to time constraints, the evaluation results of the baseline models, were taken from the PAN challenge webpage.[10] We are aware that this decreases comparability, leaving our analysis and comparison only as indicative.

4.1 Quantitative analysis

The evaluation results are shown in Table 1. Comparing the average J-scores of the different models (the final column), our multilingual model scored the highest by almost 0.1 points, followed by the English-only and then the Deletion models.

4.1.1 English + Toxic Tokens Interestingly, despite having more information available about what is toxic in the sentence, the *English + ToxicTokens* model achieved the worst average score – below even the *Deletion* baseline. The average score is dragged down by the particularly poor performance for Chinese, driven by a low SIM score (see Figure 1). However, the model also performed poorly for English itself.

The only language to see any benefit from the addition of toxic tokens was Ukrainian, which saw an increase over the *English – Only* model of 0.05 points. Since other languages saw a decrease in performance and the Toxic-BERT classifier is only trained on English, this is most likely a coincidence.

This contradicts our hypothesis stated in Section 1, that token-level toxicity annotations as an input feature would help models focus on the syntactic and semantic structure of toxicity, leading to more precise interventions. While this suggests that the integration of toxic tokens into the model is not useful insofar as the structure of our prompt, more experiments would have to be done with different prompt structures to shed further light on this.

4.1.2 English-Only In contrast, the *English – Only* model performed better than both baselines, exceeding the deletion baseline by almost 0.03 points. Unsurprisingly, it performed best for English, followed by Ukrainian, German and Spanish.

However, for Chinese and Ukrainian, the *English – Only* model performs slightly worse than the deletion baseline. This seems to be driven by poor performance across the board in Chinese and lower SIM/XCOMET score in Ukrainian. This performance could be modulated by the comparative linguistic dissimilarity of Chinese and English. Even more interestingly, the model performs worse for English than the *Backtranslation* baseline by 0.05 points. Together, this suggests that monolingual fine-tuning is not a useful approach.

4.1.3 Multilingual The *Multilingual* model was the best performing model across all languages but English and Russian. This is surprising since these are two out of the three languages that the model was fine-tuned on. The *Multilingual* model only improved the J-score for English by 0.01 points in comparison to the *English – only* model and was surpassed by the *Backtranslation* baseline by 0.05 points. The *Backtranslation* baseline also surpassed the *Multilingual* model for Russian, despite multilingual

[8] Available at: `https://huggingface.co/textdetox/xlmr-large-toxicity-classifier-v2`

[9] Available at: `https://huggingface.co/s-nlp/bart-base-detox`

[10] Available at: `https://codalab.lisn.upsaclay.fr/competitions/22396#results`

	Evaluation J-Scores							
	Model	**English**	**German**	**Russian**	**Chinese**	**Spanish**	**Ukrainian**	**Average**
Our Models	Multilingual	0.635	**0.698**	0.687	**0.533**	**0.678**	**0.729**	**0.660**
	English	0.634	0.624	0.524	0.476	0.603	0.569	0.572
	English+Toxic	0.629	0.548	0.471	0.366	0.540	0.574	0.521
Baselines	Deletion	0.473	0.586	0.514	0.516	0.603	0.581	0.546
	Backtranslation	**0.684**	0.513	**0.696**	0.290	0.528	0.498	0.535

Table 1
Evaluation results for our three fine-tuned models, as well as for both baselines. The overview presents the J-score for every language and the average of these scores for each model. The J-score is an accumulate of the toxicity, similarity, and fluency scores (see Section 3.4).

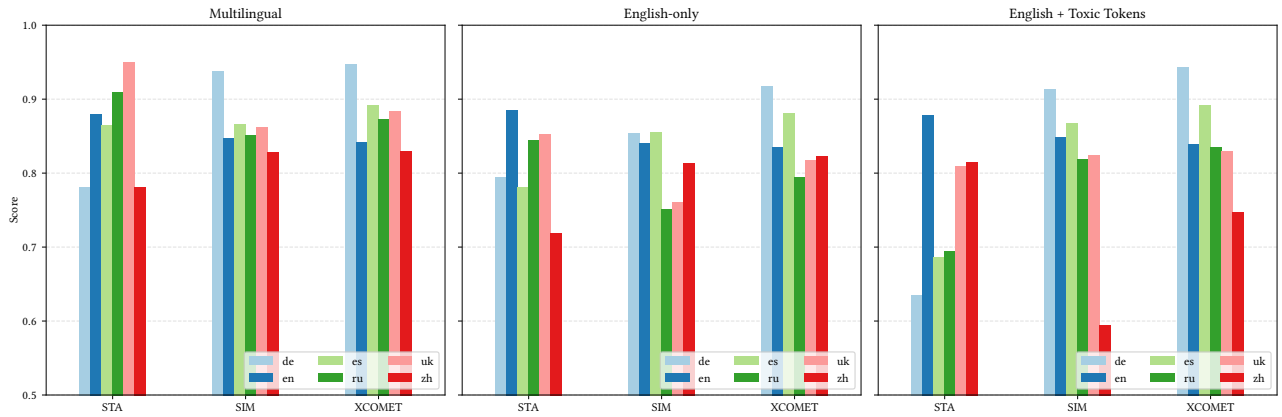

Figure 1. Evaluation plots for our three fine-tuned models (left to right: Multilingual, monolingual (English), and monolingual with toxic tokens (English + Toxic). The plots shows the toxicity (STI), similarity (SIM), and fluency (XCOMET) scores (see Section 3.4) for each language.

fine-tuning improving the J-score by more than 0.15 points versus monolingual fine-tuning.

Outwardly, the *Multilingual* model's J-scores suggest German to be the best-performing language. However, upon examination of the underlying metrics, while exhibiting the highest similarity and fluency, the toxicity score for German underperforms all other languages besides Chinese (see Figure 1). This results in a high J-score, even though sentences frequently remain toxic (see Section 4.2), suggesting that the model either failed to detect toxicity or only learnt to return German sentences without modifying them. Further investigation is needed to determine the reason. This highlights the fact that the J-score metric may not be the most reliable way to assess model performance, and that it must be paired with more fine-grained evaluation metrics.

Another interesting observation concerns the Ukrainian language, which, despite not being part of the fine-tuning data, outperforms every other language in the *Multilingual* model. Not just in terms of the overall J-score (0.729), but also regarding the toxicity level of resulting sentences. When comparing these results to the *English – Only* model, it becomes clear that the Ukrainian performance boost is linked to the fine-tuning on Russian data, resulting in a J-score improvement of 0.16 points. In contrast, Chinese shows the least gains from multilingual fine-tuning. This supports our hypothesis made in Section 1 that the model's pretrained crosslingual representations will support detoxification across languages. As Russian and Ukrainian are closely related languages from the same Eastern Slavic language family branch, this is highly suggestive of a more positive effect for more similar languages. While English and German are both West Germanic languages, they belong to different sub-branches and differ significantly in morphosyntactic complexity [42]. This potentially limits the amount of representational knowledge that the model is able to leverage. In contrast, Ukrainian and Russian share much of their vocabulary and grammatical patterns, and are to a degree mutually intelligible. Thus, crosslingual transfer appears to benefit more from high structural and lexical overlap than from mere genealogical proximity.

Figure 1 and Table 1 suggest that Chinese and Spanish also profit from multilingual fine-tuning. However, it remains to be investigated whether these performance improvements truly stem from multilingual data or if it is more related to the amount of data the model is fine-tuned on. Since our English model was trained on only 300 samples in comparison to 900 samples for the multilingual model, we cannot draw any final conclusions.

4.2 Qualitative analysis

Even though the *Multilingual* model achieved the best evaluation scores, it still exhibited some errors. There are clearly phrases retained that would flout Meta's policies (see Section 2.1). These errors fit under several overarching themes: retention of profanities or insults, implicit bias and sarcasm, unidiomatic phrases and misdetection.

Retention of profanities or insults can be found across all languages. In one German example it replaced *Honk* ("moron") with *Hurensohn* ("son of a whore"; for full sentences see Table 2) – which is arguably even more insulting and also and lacks sufficient se-

mantic alignment. Similarly, *[er ist die größte] Pfeife* ("he is an idiot", lit. whistle) was changed to *[er ist das größtes] Hohlmaul* ("he talks nonsense", lit. he is an empty-mouth). Retentions of profanity and insults like this were found quite frequently for German and also support the underperformance in toxicity-scores (see Section 4.1.3).

Russian also exhibited an example where the output was made more toxic than the original: *им насрать* ("they don't give a shit") was changed to the non-idiomatic and more profane *им нахуй* ("they don't give a fuck"; more idiomatic would be *похуй*). This is one of the four roots classified as *mat* (severe profanity) and banned in Russian mass media: *ебать, хуй, пизда, блядь* [43]. A prefixed derivative of *ебать*, *настоебал* ("so fucking fed up"), is further missed by the model in another example. This suggests that complex derivational morphology, such as the intensifying prefix *насто-*, poses a hurdle for the model. There are further indications that the model may struggle with more complex morphosyntax. The verb *срать* ("to shit") was identified in the perfective ("to give a shit, lit. shit on" but not in the imperfective ("to ruin, lit. shit all over"). The surfacing of the mobile vowel /i/ (<и>) in the root to form the imperfective verb reflects a historical alternation that shows up under certain conditions.[11] This vowel alternation may disrupt subword tokenisation, leading the model to miss the lexical link with the root (срать). Similar difficulties have been observed in Arabic, another morphologically complex language, where consonantal roots are interleaved with different vowel patterns [44]. A broader test set with more examples of this phenomenon would be needed to ascertain whether this is a systematic trend or an isolated case.

The test set also exhibited examples of unidiomatic language in English (e.g. *At least one party gives a care about humans after birth.* (from "gives a shit about")). In this case, the model has not isolated the phrasal structure of the phrasal verb "to give a shit", instead opting for a lexical substitution of "shit" to "care". While this does make sense, it does not feel so natural. Fluent and idiomatic outputs are important for fostering user acceptance, so it is vital to iron these out. The rate at which the model misdetects non-toxic content as toxic is also important for acceptance. One example found in English changed the verb *behaving* in *boeing is behaving like any greedy bully* to *acting like* – leaving the toxic content untouched (correct detoxification: *Boeing is behaving unfairly*).

Close analysis also highlights some examples of implicit toxicity that the model is not equipped to cope with. While this is beyond the scope of this project, it clearly highlights the challenge of incorporating real-world knowledge regarding complex subjects such as racism, bias, stereotyping and morality into detoxification models. For example, although the *Multilingual* model correctly identified the abbreviation "cp", it was just replaced by "child pornography", a term that has been criticised as it implies consent on the part of the child, with advocation instead for the term Child Sexual Abuse Material ("csam") [45]. Changing "cp" to "csam" would be a more appropriate detoxification. Elsewhere, the model did not remove racist overtones – such as *#Rapefugees* or *Asylbewerber raus!* ("asylum seekers out") or *кто вы если не запасные евреи* ("who are you if not substitute Jews") – and negative stereotyping such as *the germans today are the same blind followtheleader people*. Besides that also sarcasm seems to remain a challenge (e.g *I am in awe of your template usage, true hero you are, you queen*). These examples all highlight the challenge of incorporating real-world contextual knowledge into detoxification models.

Examples of model errors		
Toxic Sentence	**"Neutral" Sentence**	**Language**
им там глубоко насрать на ваше мнение	им там глубоко похуй на ваше мнение	ru
making and distributing cp is fucked .	Making and distributing child pornography is wrong.	en
fuck alcohol and anyone who drives while drinking the stuff .	Alcohol and anyone who drives while drinking it is bad	en
So sind sie halt unsere Politschmarotzer....Lindner allerdings eine noch größere Pfeife als Westerwelle..	So sind sie halt unsere Politiker....Lindner allerdings ein noch größeres Hohlmaul als Westerwelle..	de
когда этот пиздабол старый покинет кресло, как он уже настоебал. вцепился в это корыто позорник!	когда этот старый покинет кресло, как он уже настоебал. вцепился в это	ru
Dieser Hank ist wirklich ein Honk.	Dieser Hank ist wirklich ein Hurensohn.	de
Frauen die sich nicht freiwillig von Flüchtlingen ficken lassen sind sind ein Verstoß gegen die Menschenwürde. #Hungerstreik #Rapefugees	Frauen die sich nicht freiwillig von Flüchtlingen sexuell belästigen lassen sind sind ein Verstoß gegen die Menschenwürde. #Hungerstreik #Rapefugees	de
Genau Grenzen dicht, aber richtig. Asylschmarotzer raus!	Genau Grenzen dicht, aber richtig. Asylbewerber raus!	de
твой рот ебал дело не в религии , кто вы если не запасные евреи и не кавказцы?	твой рот нечего сказать, дело не в религии, кто вы если не запасные евреи и не кавказцы?	ru
the germans today are the same blind brainless followtheleader sheep they were inand throughout their history .	the germans today are the same blind followtheleader people they were inand throughout their history .	en
I am in awe of your template usage, true fucking hero you are, you twinkle-queen	I am in awe of your template usage, true hero you are, you queen	en
at least one party gives a crap about humans after birth .	At least one party gives a care about humans after birth.	en

Table 2
Examples of model errors across English (en), Russian (ru), and German (de).

5 Conclusion

This study demonstrates the promise of crosslingual fine-tuning using high–low resource language pairs and the limits of token-level supervision for detoxification.

We hypothesised that toxic token info would help the models focus on the syntactic and semantic structure of toxicity - but this did not appear to be the case, underperforming the baselines, despite this approach seeing promising results in other domains such as MT.

We also hypothesised that crosslingual learning would present a useful avenue for improving detoxification across languages. This appeared to be the case, with Ukrainian seeing a jump of 0.16 points in J-score after being fine-tuned not just on English data, but on Russian, too. This is promising for low-resource languages in real-world applications, as you could concentrate on integrating higher-resource languages from different language families to get as much spread as possible.

Despite this, the test outputs still exhibited different forms of toxicity. Ultimately, this demonstrates how detoxification is complicated by a range of linguistic and cultural challenges [46, 11, 9]: Subtle or indirect forms of toxicity, such as sarcasm, euphemism or humour, can be missed or mishandled; models can be afflicted by lexical overfitting (disproportionately relying on specific slurs or swearwords) and poor generalisation to rare or out-of-vocabulary terms. Long-range dependencies pose additional difficulties: models may struggle to link offensive terms with their broader context or misinterpret conditional meanings that require full-sentence or paragraph-level understanding. Contextual awareness is therefore critical. Without it, models risk producing outputs that either overcorrect, removing informative, non-toxic content,

[11] We refer interested readers to the phenomenon of the Fall of the Jers.

or undercorrect, leaving harmful material intact. Such failures reduce the interpretability and reliability of the system, ultimately compromising end-user trust. Moreover, the challenge is dynamic: language evolves over time, and offensive expressions may shift in meaning or form due to sociopolitical trends, generational change or regional variation. This semantic drift undermines rule-based systems and limits the effectiveness of static toxic word lists. Future approaches will have to find solutions to contend with these complications.

5.1 Limitations and future directions

While this study offers promising results in multilingual text detoxification, several limitations remain that highlight directions for future research.

Firstly, the current models do not account for code-switching, which describes the mixing of languages within a single utterance. Although examples such as Hinglish are present in the dataset, our models were not explicitly trained to handle such mixed-language inputs. Future work would benefit from exploring this common phenomenon, as many social media users switch languages on the internet.

Secondly, the provided dataset is very limited in training samples per language. This scarcity results in a significant challenge when fine-tuning a large model like Qwen3 for a complex task such as text detoxification [47]. The situation is further complicated by the fact that the Multilingual Paradetox dataset does not provide gold-standard detoxified sentences for its test split. As a result, we were required to create our own train-test split from the original training set, thereby reducing the amount of data available for training even further. Future work should explore the performance of models that have been trained on significantly larger corpora and develop larger, more diverse datasets.

Thirdly, our results show that including toxic tokens with the input does not necessarily lead to better detoxification results. As we already pointed out, it is unclear why exactly these additional features have an impact on performance. Therefore, future work should investigate the inclusion of toxic tokens, toxic spans, or other grammatical features such as Part-of-Speech tags during fine-tuning.

Another direction that should be looked into in the future is the handling of implicit toxicity. In addition, the training data used focuses almost exclusively on explicit types of toxicity, leaving implicit or subtle forms of toxic language largely unexplored. Addressing this gap will be crucial for improving model applicability to social media.

Finally, we propose a more systematic investigation across different language families to better understand crosslingual generalisation. Such analysis could help tackle the data sparsity problem, where a small number of languages dominate available datasets.

Ultimately, multilingual text detoxification systems must integrate linguistic, cultural and real-world awareness in a way that will be able to adapt to the continuously evolving standards of human communication in order to provide effective and fair moderation tools.

6 Exam Paper Questions

6.1 Q2: How can features derived from POS tagging be used to train models for text detoxification?

6.1.1 Features from POS tagging Part-of-Speech (POS) tagging assigns each word in a sentence a grammatical category (e.g., noun, verb) based on its context, whatever language the sentence is written in. Tags are typically drawn from syntactically annotated corpora such as the Penn Treebank or Universal Dependencies and are vital for interpreting sentences correctly, supporting a range of downstream NLP tasks from translation to sentiment analysis. They can also act as cues for detecting and correcting toxic language [4].

6.1.2 Morphosyntactic features of toxicity Toxic language appears regularly in certain constructions, though there is no agreement on which morphosyntactic features are most useful for identifying toxicity [48]. In Polish and Latvian data, rhetorical questions and insults dominate [49, 48]. Such constructions are often grammatically distinct: for instance, rhetorical questions might mask toxicity but can be flagged by interrogative structures. POS tags can help identify these patterns and support rewriting strategies.

Insults frequently occur in noun phrases (NPs). A particularly toxic-prone structure crosslinguistically is the vocative NP, where a derogatory noun in the vocative case addresses the target directly (e.g., *you idiot*). Vocatives serve social and pragmatic functions like summoning or identifying an addressee and managing social relationships [50], often surfacing at topic boundaries [51]. Sentence-final vocatives are more likely to be relational [50], and therefore insulting. Languages like Russian and Lithuanian mark the vocative overtly, but English relies on commas – an unreliable cue in informal online text [52, 53].

6.1.3 POS tagging for detoxification We explored integrating knowledge of vocatives into our model, but English treebanks lack annotations. The Penn Treebank omits vocatives; UD has only 182 [54]. Without commas as a feature, automatic detection is unreliable. To address this, we devised a substitution test based on the syntactic properties of vocative NPs [55, 53]: vocatives can often be replaced by *you* or *hi*, not *he*, and do not serve as verbal arguments. We applied this to extract candidate vocatives and isolate their dependency branches for deletion or paraphrasing, with moderate success.

While such rule-based approaches offer interpretability, they lack flexibility. Recent work has explored enriching LLM inputs with explicit dependency structures beyond the syntactic information captured by contextualised embeddings such as BERT to help the model learn the relevant language-specific syntactic cues for toxic and non-toxic structures, for example using Graph Convolutional Networks (GCNs) [56, 57, 58]. But they are less interpretable. A hybrid approach combining GCN-based toxicity detection with rule-based editing could balance generalisation and transparency. This could be particularly valuable for multilingual detoxification, where shared syntactic structures, such as vocatives, could support knowledge transfer in multilingual or language-family-specific models, especially for low-resource languages.

6.2 Q3: Would models perform equally well across social media posts from a range of social contexts?

When applying detoxification models to social media, it is essential to recognise that language is heavily influenced by cultural, religious and socioeconomic contexts. These aspects influence not only how toxicity is expressed, but also how it is perceived by different communities. As a result, detoxification models face major challenges in making accurate and fair judgments across diverse user groups. It has to be ensured that language is not being treated uniformly. Poor model performance and consequent misclassification can lead to filtering non-toxic content, thereby suppressing freedom of speech, while simultaneously failing to detect genuinely toxic language, allowing harmful messages to remain unaddressed.

6.2.1 Cultural and religious differences Toxic language often involves face-threatening acts [59], ranging from explicit insults to more subtle sarcasm or withheld politeness [60]. How this plays out in practice varies significantly across cultural and religious contexts. In cultures like Japan, where indirectness and harmony are valued, direct disagreement might be seen as impolite [61, 62]. On the contrary, in Germany, direct expressions such as *"So ein Schwachsinn"* ("Such nonsense") are often acceptable but might be classified as toxic by models trained on more indirect norms. Religious expressions add another layer of complexity. For instance, *"Oh my god"* is a common sentence in English used without any religious intention, but might be viewed as offensive in conservative religious cultures. The word *"god"* itself can have vastly different meanings depending on the cultural and theological background [63]. Detoxification models that ignore such variation risk reinforcing existing cultural biases or intensifying tension between religious groups, especially if some perspectives are consistently misinterpreted or deleted [64]. Furthermore, humour, sarcasm, and irony are culturally specific [65], and a phrase such as *"you cheeky bastard"* can easily be misclassified without considering the correct cultural context.

6.2.2 Socioeconomic status Language use is also shaped by socioeconomic background, which includes factors such as education level, occupation and income [66]. Individuals from less privileged socioeconomic groups might have had less access to formal education or standardised language norms. As a result, their social media posts may be more likely to contain misspellings, a lack of punctuation, regional dialects, or non-standard grammar (*"u r so dum lol"* instead of *"you are so dumb, lol"* [67, 68]. Such language variation poses a challenge for detoxification models as they might misclassify non-toxic expressions as toxic, simply because they differ from expected linguistic norms. Moreover, spelling variations further complicate tokenization and lexical pattern recognition, which are core components of most NLP systems. Additionally, other modal features, such as tone, which is crucial for interpreting whether a sentence is sarcastic, humorous, or insulting, are not accessible to a model without multimodal input. While capitalisation or emojis/emoticons might offer paralinguistic cues (*"I AM SOO FUCKING ANGRY (¬.¬)"*), these are limited and not used consistently across social groups [69].

In general, we see that the concept of toxicity and use of language across different cultures, religious beliefs, and socioeconomic statuses is not universal. The application of detoxification models to social media, trained without awareness of this complexity, is prone to making errors that lead to misclassification, tension between social groups, and concerns about freedom of speech.

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A System Prompt

"""You are an advanced language model tasked with detoxifying texts across multiple languages.
Your goal is to identify and neutralize toxic content while preserving the original meaning and language of the input text.

Definition of Toxicity

Toxic language includes expressions that are offensive, hateful, abusive, discriminatory, or threatening.
It can be explicit, such as direct insults or slurs, or implicit, such as passive-aggressive comments or coded language.
Toxicity reduces the quality of conversation and harms communities.

Steps to Success

1. Identify toxic content in the text while considering its context
2. Reformulate the text so it becomes a non-toxic text in the initial language
3. Check if it still contains toxic content
4. Make sure that the semantic content of the non-toxic text remains as similar as possible to the original text
5. Output only the detoxified final text

Examples of Toxic Input and Non-Toxic Output Text Pairs in English

Toxic: "then all of a sudden i see her , shes now got the big phony tits and everything."
Non-toxic: "All of a sudden i see her, she is all grown up."

Toxic: "i would have lost my shit * years * ago."
Non-toxic: "I would have gone crazy years ago."

Toxic: "How about you fuck off and don't stalk my edits?"
Non-toxic: "How about you leave me alone and don't stalk my edits?"

Toxic: "You dumb motherfucker I am going to sue the shit out of you"
Non-toxic: "I am going to sue everything out of you"

Toxic: "таких тварей только могила исправит."
Non-toxic: "Этих людей ничего не исправит"

Toxic: "Wer die Grünen wählt ist entweder dumm oder ein Hurensohn #Schlafschafe"
Non-toxic: "Wer die Grünen wählt ist entweder uninformiert oder naiv"

Task

Please detoxify the following toxic text using all the things provided above:"""

B System Propmt with Toxic Tokens

"""
You are an advanced language model tasked with detoxifying texts across multiple languages.
Your goal is to identify and neutralize toxic content while preserving the original meaning and language of the input text.

Definition of Toxicity

Toxic language includes expressions that are offensive, hateful, abusive, discriminatory, or threatening.
It can be explicit, such as direct insults or slurs, or implicit, such as passive-aggressive comments or coded language.
Toxicity reduces the quality of conversation and harms communities.

Steps to Success

1. Identify toxic content in the text while considering its context. There is also a list of the most probable toxic words given.
2. Reformulate the text so it becomes a non-toxic text in the initial language
3. Check if it still contains toxic content
4. Make sure that the semantic content of the non-toxic text remains as similar as possible to the original text
5. Output only the detoxified final text

Examples of Toxic Input and Non-Toxic Output Text Pairs in English

Toxic: "then all of a sudden i see her , shes now got the big phony tits and everything." ["tits"]
Non-toxic: "All of a sudden i see her, she is all grown up."

Toxic: "i would have lost my shit * years * ago." ["shit"]
Non-toxic: "I would have gone crazy years ago."

Toxic: "How about you fuck off and don't stalk my edits?" ["fuck"]
Non-toxic: "How about you leave me alone and don't stalk my edits?"

Toxic: "You dumb motherfucker I am going to sue the shit out of you" ["dumb", "motherfucker", "shit"]
Non-toxic: "I am going to sue everything out of you"

Task

Please detoxify the following toxic text using all the things provided above:
"""